

# Moein Razavi, Ph.D.

## Generative AI Engineer | Agentic Systems, RAG & Production AI on Google Cloud

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### SUMMARY

Hands-on generative AI engineer with a Ph.D. and 8+ years of Python software development, taking agentic AI from prototype to production on Google Cloud. At Ford, architects, codes, and launches agentic systems for business teams: multi-agent assistants with routing, hierarchical delegation, and checker agents; RAG and text-to-SQL over structured and unstructured data; and Gemini-powered ingestion pipelines that connect messy third-party data to BigQuery, cutting manual work by 30–60%. Deploys on Vertex AI and Cloud Run with Terraform and CI/CD, building security boundaries (SAML/ADFS SSO, PII redaction), evaluation, and validation into every system. As a founder and an embedded contract engineer, has run requirements discovery directly with customers and shipped what they needed.

### CORE QUALIFICATIONS

- **Production AI on GCP, conception to launch:** Vertex AI, Gemini, Cloud Run, BigQuery, Dataproc, Cloud Storage, Terraform, Docker, Tekton and GitHub Actions CI/CD.
- **Agentic systems:** Google ADK, LangGraph, LangChain, Model Context Protocol (MCP) servers; intent routing, hierarchical delegation, self-reflection via checker agents and validate-and-retry loops, human-in-the-loop approval.
- **Data for enterprise AI:** RAG over unstructured documents, schema-aware text-to-SQL over structured data, BM25 and vector search (FAISS, ChromaDB), LLM-driven schema discovery and extraction from PDF, HTML, XML, and spreadsheets.
- **Evaluation and reliability:** deterministic output validation, domain-rule answer checks, grouped cross-validation, bootstrap confidence intervals, backtesting against baselines, cost estimation for Vertex AI workloads.

### PROFESSIONAL EXPERIENCE

#### Ford Motor Company

Aug 2023 – Present

*Senior Data Scientist (Generative AI and Machine Learning) | Remote*

- **Multi-agent analytics assistant, prototype to production:** Cut time spent answering recurring analytics and document questions by **30%**. An intent router delegates each request to a schema-aware text-to-SQL agent over BigQuery or a RAG agent over legal documents and vehicle manuals, and a checker agent reviews every answer against domain rules before it reaches the user. Built with LangChain and GPT-4o; launched on Cloud Run behind SAML/ADFS single sign-on.
- **Agentic ingestion of unstructured third-party data (HOMELOAD):** Reduced manual data preparation by **~50%** with a 5-agent Python/FastAPI service that connects inconsistent utility-bill data (CSV, Excel, PDF, XML, HTML) to BigQuery. Gemini performs schema discovery and extraction planning; PII redaction, deterministic validation, bounded self-correcting retries, and idempotent BigQuery MERGE loads make it production-safe. Deployed with Docker, Cloud Run, and Terraform.
- **LLM developer tooling:** Reduced manual code-review time by **60%** with a FastAPI service that parses repositories (GitPython, Python AST), streams LLM-assisted complexity reviews to the browser over Server-Sent Events, and estimates the Vertex AI cost of running the code.
- **Transformer forecasting on Vertex AI:** Reduced vehicle-availability forecast error by **~14%** versus the previous baseline across **~9,500 vehicles** with PatchTST-style transformers in PyTorch, benchmarked against TimesFM, MOIRAI-2, and iTransformer and tuned with Optuna on Vertex AI GPUs; Monte Carlo capacity estimates feed ERCOT electricity-market bids.
- **Data foundations for AI:** Build and maintain EV telemetry pipelines processing **10+ TB** across **15 Airflow DAGs** (PySpark on Dataproc, BigQuery, Dataform) and debug production data issues end to end; delivered XGBoost bill-prediction APIs (Flask, BigQuery) with a SAML-authenticated Dash front end through Tekton CI/CD.

#### CereVu Medical, Inc.

Jun 2023 – Sep 2024

*Software Developer / Research Engineer (Embedded Contract Engineer) | Remote*

- Embedded with a medical-device startup's team: turned clinician and founder requirements into software, then developed and launched **two SwiftUI iOS apps** capturing **9 physiological signals** from wearable sensors over Bluetooth Low Energy, through TestFlight and App Store submissions.
- Built the connective layer from device to clinician: Python/Flask services with WebSocket streaming and MongoDB feeding a React/Next.js dashboard, OAuth 2.0/JWT authentication with role-based access, and Docker services on AWS EC2 with GitHub Actions CI/CD.

- **Rigorous model evaluation (first author):** Built the evaluation pipeline for wearable stress detection on Apple Watch data from 50+ participants: participant-grouped cross-validation, bootstrap confidence intervals, permutation tests, fixed false-positive-rate operating points, and SHAP explanations across 6 model families.
- **MERCURY (IEEE HPCA 2023, co-author):** Accelerated deep neural network training with random projection, quantization, and cached computation reuse; reported **1.97x** average speedup across 12 models at baseline-level accuracy.
- **Computer vision (first author):** Fine-tuned TensorFlow Faster R-CNN detectors to **99.8%** face-mask classification accuracy on a 3,300-image dataset and measured worker distancing in construction video; built OpenSync, an open-source Python sensor-synchronization platform (sub-0.07 ms RMS lag) released on PyPI.

## SELECTED PROJECTS

- **Business Action Prioritizer (Google ADK, Gemini, MCP):** Built a 4-stage agent application that turns business spreadsheets and a scoring rubric into ranked actions. A planning agent proposes the analysis and pauses for human approval; an execution agent tracks plan state and delegates to a PandasAI child agent; an MCP server delivers reports through signed Cloud Storage URLs; FastAPI and SQLite persist sessions.
- **Inspealytics (Founder & Technical Lead, 2023 – Present):** Ran discovery with inspection clients, then designed and launched a product that cut inspection-report preparation time by **~50%**: SwiftUI iOS app, React/TypeScript web app, Supabase (PostgreSQL), offline edit conflict resolution, Apple Vision photo organization, automated Word/PowerPoint reports, and XCTest/Vitest test suites.
- **On-device transformer (mHELP, 2026):** Deployed a PyTorch-trained transformer to Apple Watch through Core ML, reimplementing training-time preprocessing in Swift so on-device predictions match the trained model.

## TECHNICAL SKILLS

**Languages:** Python (8+ years), SQL, TypeScript/JavaScript, Swift, C++, Java

**Generative AI:** Gemini, GPT-4o, Claude; Google ADK, LangGraph, LangChain, MCP, PandasAI; RAG, text-to-SQL, tool calling, structured outputs, prompt design

**Google Cloud:** Vertex AI, Cloud Run, BigQuery, Dataproc, Dataform, Cloud Storage; also AWS (EC2, S3, Lambda)

**Machine Learning:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, Hugging Face, Optuna, transformers, time-series forecasting, computer vision, SHAP

**Engineering:** FastAPI, Flask, REST, WebSocket, Server-Sent Events, PySpark, Airflow, PostgreSQL, MongoDB, Docker, Kubernetes, Terraform, Tekton, GitHub Actions, OAuth 2.0, JWT, SAML/ADFS, React/Next.js, SwiftUI

## EDUCATION

**Ph.D., Industrial Engineering (Data Science)** | Texas A&M University | GPA 4.0/4.0 Dec 2023

**Master of Engineering, Computer Engineering** | Texas A&M University | GPA 4.0/4.0 May 2021

**B.S., Mechanical Engineering** | Amirkabir University of Technology (Tehran Polytechnic) Jun 2018

## SELECTED PUBLICATIONS

Janfaza, V., Weston, K., **Razavi, M.**, et al. MERCURY: Accelerating DNN Training by Exploiting Input Similarity. *IEEE HPCA*, 2023.

**Razavi, M.**, et al. Machine Learning, Deep Learning, and Data Preprocessing Techniques for Detecting, Predicting, and Monitoring Stress and Stress-Related Mental Disorders: Scoping Review. *JMIR Mental Health*, 2024.

**Razavi, M.**, et al. OpenSync: An Open-Source Platform for Synchronizing Multiple Measures in Neuroscience Experiments. *Journal of Neuroscience Methods*, 2022.